WORKED SOLUTION

STEP III 2007, Question 12

Probability and Statistics • Random Sums

STEP III 2007, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q12

The key idea

Y is a sum with a *random* number of terms, so we can't use the rules for sums of a fixed number of variables straight away. The trick is to condition on N: once we know $N = k$, Y is an ordinary sum of k independent variables, whose mean and variance we know. Then we average over the distribution of N, which is why the first part asks for $\mathrm{E}(N)$ and $\mathrm{E}(N^2)$.

Solution

The distribution of N

We take the number to be chosen uniformly at random, so $\mathrm{P}(N = k) = \frac{1}{2n-1}$ for $k = 1, 2, \ldots, 2n-1$. Using $\sum_{k=1}^{m} k = \frac{1}{2}m(m+1)$ and $\sum_{k=1}^{m} k^2 = \frac{1}{6}m(m+1)(2m+1)$ with $m = 2n-1$:

$$\mathrm{E}(N) = \frac{1}{2n-1} \cdot \frac{(2n-1)(2n)}{2} = n,$$
$$\mathrm{E}(N^2) = \frac{1}{2n-1} \cdot \frac{(2n-1)(2n)(4n-1)}{6} = \frac{n(4n-1)}{3}.$$

(The mean n is also clear by symmetry: n is the middle of $1, \ldots, 2n-1$.) For later use,

$$\mathrm{Var}(N) = \frac{n(4n-1)}{3} - n^2 = \frac{n^2 - n}{3} = \frac{n(n-1)}{3}.$$

Conditioning on N

Fix k and suppose $N = k$. Because the X_i are independent of N, conditioning on $N = k$ doesn't change their distributions, so given $N = k$ we have $Y = X_1 + \cdots + X_k$, a sum of k independent variables each with mean μ and variance σ^2. Hence

$$\mathrm{E}(Y \mid N = k) = k\mu, \qquad \mathrm{Var}(Y \mid N = k) = k\sigma^2,$$

the second using independence of the X_i (variances add). So

$$\mathrm{E}(Y^2 \mid N=k) = \mathrm{Var}(Y \mid N=k) + (\mathrm{E}(Y \mid N=k))^2 = k\sigma^2 + k^2\mu^2.$$

Ask yourself How do we get from a conditional expectation back to an ordinary one?

By the law of total expectation (the partition theorem): $\mathrm{E}(g(Y,N)) = \sum_k \mathrm{P}(N=k)\,\mathrm{E}(g(Y,N) \mid N=k)$. Inside each term N is just the number k.

The mean of Y

$$\mathrm{E}(Y) = \sum_k \mathrm{P}(N=k)\,k\mu = \mu\,\mathrm{E}(N) = n\mu,$$

as required.

The covariance

Given $N=k$, $YN = kY$, so $\mathrm{E}(YN \mid N=k) = k \cdot k\mu = k^2\mu$. Therefore $\mathrm{E}(YN) = \mu\,\mathrm{E}(N^2)$ and

$$\begin{aligned}\mathrm{Cov}(Y,N) &= \mathrm{E}(YN) - \mathrm{E}(Y)\,\mathrm{E}(N) = \mu\,\mathrm{E}(N^2) - n\mu \cdot n \\ &= \mu\,\mathrm{Var}(N) = \tfrac{1}{3}n(n-1)\mu,\end{aligned}$$

which is the required result.

The variance of Y

Averaging $\mathrm{E}(Y^2 \mid N=k) = k\sigma^2 + k^2\mu^2$ over N:

$$\mathrm{E}(Y^2) = \sigma^2\,\mathrm{E}(N) + \mu^2\,\mathrm{E}(N^2) = n\sigma^2 + \frac{n(4n-1)}{3}\mu^2.$$

Hence

$$\mathrm{Var}(Y) = n\sigma^2 + \frac{n(4n-1)}{3}\mu^2 - n^2\mu^2 = n\sigma^2 + \frac{n(n-1)}{3}\mu^2.$$

The two terms have a natural meaning: $n\sigma^2$ is the average spread of Y for a fixed number of trials, and $\frac{1}{3}n(n-1)\mu^2 = \mu^2\,\mathrm{Var}(N)$ is the extra spread caused by not knowing how many trials there will be.

Checks

- If $n=1$ then $N=1$ always, and the formulas reduce to $\mathrm{E}(Y)=\mu$, $\mathrm{Cov}(Y,N)=0$ and $\mathrm{Var}(Y)=\sigma^2$, as they should for a single X_1.
- If $\mu=0$ then the extra term vanishes: the number of terms no longer affects the mean, so it can't add spread or correlate with Y.

- Take $n = 2$ and each X_i equal to 0 or 1 with probability $\frac{1}{2}$. Then $N \in \{1, 2, 3\}$ and $Y \mid N \sim \mathrm{B}(N, \frac{1}{2})$; the formula gives $\mathrm{Var}(Y) = 2 \cdot \frac{1}{4} + \frac{2}{3} \cdot \frac{1}{4} = \frac{2}{3}$. We confirmed this, and all the results, by exact enumeration of the joint distribution of (N, Y) for $n = 1, \ldots, 6$ and several distributions of X_i.

Common slips

- Writing $\mathrm{Var}(Y) = \mathrm{E}(N)\sigma^2$ only, forgetting that N itself varies. The second term is the whole point.
- Treating Y and N as independent and getting $\mathrm{Cov}(Y, N) = 0$. More trials means a bigger sum when $\mu > 0$, so they are positively correlated.

Beyond STEP

Y is a *random sum* (a compound distribution), and the variance result is a case of the law of total variance, $\mathrm{Var}(Y) = \mathrm{E}(\mathrm{Var}(Y \mid N)) + \mathrm{Var}(\mathrm{E}(Y \mid N))$, here $\sigma^2\mathrm{E}(N) + \mu^2\mathrm{Var}(N)$. Random sums model insurance claims and branching processes, and are handled elegantly with probability generating functions: $G_Y(s) = G_N(G_X(s))$. Grimmett and Welsh, *Probability: An Introduction*, treats random sums and branching processes this way.

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